# BitLocker Drive Encryption

Implementation Solution Guide

Version 1.0.0 · 2026-06-13

# Contents


Executive Summary — BitLocker Drive Encryption (Microsoft Intune) 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 4

Outcome . . . 4

1. Solution Overview — BitLocker Drive Encryption 4

What BitLocker is . . . 4

How it works . . . 5

What you can require . . . 5

Benefits . . . 5

Limitations (be honest with buyers) . . . 5

The standard deployment model . . . 5

2. Requirements Guide — BitLocker Drive Encryption 5

2.1 Licensing . . . 6

2.2 Windows edition . . . 6

2.3 TPM (for silent enablement) . . . 6

2.4 UEFI / Secure Boot (recommended) . . . 6

2.5 Microsoft Entra ID (key escrow) . . . 6

2.6 Device enrollment . . . 6

Checklist . . . 6

Global Requirements — Intune Solutions (category-level) 7

G1. Microsoft Entra ID tenant . . . 7

G2. Microsoft Intune subscription . . . 7

G3. Administrative roles . . . 7

G4. Device enrollment in Intune . . . 7

G5. Pilot security group . . . 7

G6. Supported Windows baseline . . . 7

G7. Network access to Microsoft cloud endpoints . . . 8

3. Implementation Guide — BitLocker Drive Encryption 8

Step 1 — Create the pilot group . . . 8

Step 2 — Create the disk encryption policy . . . 8

Step 3 — Configure BitLocker settings . . . 8

Step 4 — Assign to the pilot group . . . 9

Step 5 — Apply on a pilot device . . . 9

Step 6 — Validate the pilot . . . 9

Step 7 — Expand to production . . . 9
5. Testing Guide — BitLocker Drive Encryption 9
Test 1 — Policy applied . . . 10
Test 2 — OS drive is encrypted . . . 10
Test 3 — Silent enablement (TPM devices) . . . 10
Test 4 — Recovery key escrowed to Entra . . . 10
Test 5 — Recovery key retrieval works . . . 10
Test 6 — Compliance reflects encryption . . . 10
Test 7 — Tenant report . . . 10
Pass/fail gate . . . 10
6. Troubleshooting Guide — BitLocker Drive Encryption 10
A. Encryption doesn't start . . . 11
B. Recovery key not escrowed (critical) . . . 11
C. Wrong encryption method . . . 11
D. TPM problems . . . 11
E. Diagnostics quick reference . . . 11
F. When to escalate . . . 12
7. Best Practices — BitLocker Drive Encryption 12
8. Deliverables Summary — BitLocker Drive Encryption 12
Documentation . . . 12
PowerShell scripts . . . 13
What the buyer receives . . . 13
Outcome . . . 13
Appendix A — PowerShell Scripts 14

# Executive Summary — BitLocker Drive Encryption (Microsoft Intune)

**White-label · Intune Solutions**

A laptop left in a taxi is only a breach if its drive is readable. **BitLocker** encrypts the whole disk so a lost or stolen device exposes nothing — and managed through Microsoft Intune, it turns on **silently** on TPM-equipped devices, with every **recovery key escrowed to Microsoft Entra ID** so the help desk can always recover access.

This is a complete, **deploy-it-today** package for implementing BitLocker Drive Encryption with Microsoft Intune, written for an administrator with no prior BitLocker-management experience.

## What you get

- A plain-language **Solution Overview** of how Intune-managed BitLocker works.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for readiness discovery, encryption validation, tenant reporting, remediation, and recovery-key retrieval.
- **Testing** and **Troubleshooting** guides for confident rollout.

## What this delivers

An Intune **disk encryption policy** that requires **XTS-AES 256** encryption of the OS (and fixed) drives, **silently** on TPM devices, with **recovery keys automatically saved to Entra ID** and full encryption reporting.

## The standard deployment model

**Assess → Configure → Pilot → Validate → Expand.**

## Outcome

Fleet-wide full-disk encryption with escrowed recovery keys — device-loss data protection and a compliance cornerstone — deployable by a single administrator and validated before rollout.

# 1. Solution Overview — BitLocker Drive Encryption

## What BitLocker is

BitLocker is the full-volume encryption built into Windows. It encrypts the operating-system drive (and optionally fixed data drives) so the data is unreadable without the correct key. Managed through Microsoft Intune, encryption is enforced by policy, enabled **silently** on devices with a TPM, and every **recovery key is escrowed to Microsoft Entra ID**.

## How it works

1. An Intune **disk encryption policy** defines the encryption requirements (method, drives, protectors).

2. On a targeted device, BitLocker enables — **silently** when a TPM is present and the user has no action to take.

3. The OS drive is protected by the **TPM** (optionally plus a startup PIN); the encryption uses **XTS-AES 256**.

4. The **recovery key is automatically backed up to Entra ID**, visible to admins in Intune (device → Recovery keys) and Entra (device → BitLocker keys).

5. Intune reports each device's encryption status for compliance and audits.

## What you can require

• **OS drive** encryption with TPM protector (and optional startup PIN), XTS-AES 256.

• **Fixed data drive** encryption with auto-unlock.

• **Recovery options** — store recovery key in Entra ID, hide recovery options from the user, client-driven recovery password rotation.

• **Silent enablement** for standard users on TPM devices.

## Benefits

• **Device-loss protection** — a lost/stolen device exposes no readable data.

• **Silent and zero-touch** — no user action on TPM hardware.

• **Always recoverable** — keys escrowed to Entra ID; the help desk can recover access.

• **Compliance cornerstone** — satisfies "require encryption" in device compliance.

## Limitations (be honest with buyers)

• **Silent** enablement requires a **TPM** and a supported, properly configured device.

• Devices without a TPM can encrypt but require a startup PIN/USB key (not silent).

• The first full encryption takes time (used-space-only is faster); plan pilot timing.

• Self-encrypting drives / certain firmware quirks may need hardware-specific handling.

## The standard deployment model

> ***Create the disk encryption policy → require XTS-AES 256 + Entra key escrow + silent enablement → assign to a pilot → validate encryption + key escrow → expand.*** *repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

# 2. Requirements Guide — BitLocker Drive Encryption

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

## 2.1 Licensing

**Description:** Intune (Plan 1); BitLocker is a Windows feature (no separate license). **Why:** Intune delivers and reports the disk encryption policy. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

## 2.2 Windows edition

**Description:** Windows 10/11 **Pro, Enterprise, or Education**. **Why:** BitLocker is not available on Windows Home. **Verify:** `winver` / **Settings** → **System** → **About** → **Edition**. **Implement:** Upgrade Home editions to Pro/Enterprise.

## 2.3 TPM (for silent enablement)

**Description:** TPM 1.2+ (2.0 recommended), enabled in UEFI. **Why:** Silent BitLocker enablement and the strongest protection require a TPM. **Verify:** `Get-Tpm` (`TpmPresent : True`, `TpmReady : True`); or run `BITLK-Discovery.ps1`. **Implement:** Enable TPM/PTT/fTPM in UEFI. Devices without TPM can still encrypt with a startup PIN/USB key (not silent).

## 2.4 UEFI / Secure Boot (recommended)

**Description:** UEFI firmware with Secure Boot. **Why:** Modern boot integrity improves BitLocker protection and pairs with health attestation. **Verify:** `Confirm-SecureBootUEFI`. **Implement:** Switch firmware to UEFI and enable Secure Boot (may require reinstall if legacy BIOS/MBR).

## 2.5 Microsoft Entra ID (key escrow)

**Description:** Devices Microsoft Entra joined (or hybrid joined). **Why:** Recovery keys are escrowed to the device's Entra object. **Verify:** `dsregcmd /status` → `AzureAdJoined : YES`. **Implement:** Join devices to Microsoft Entra.

## 2.6 Device enrollment

**Description:** Devices enrolled in Intune. **Why:** The policy and reporting require enrollment. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

## Checklist

- Intune licensing assigned
- Windows Pro/Enterprise/Education
- TPM present + ready (for silent) — `BITLK-Discovery.ps1`
- UEFI + Secure Boot (recommended)

- Devices Microsoft Entra joined (key escrow)
- Devices enrolled in Intune
- Pilot Entra group created

# Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

## G1. Microsoft Entra ID tenant

**Description:** An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

## G2. Microsoft Intune subscription

**Description:** An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

## G3. Administrative roles

**Description:** Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

## G4. Device enrollment in Intune

**Description:** Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

## G5. Pilot security group

**Description:** A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

## G6. Supported Windows baseline

**Description:** Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy

**Update rings** or upgrade to Windows 11.

### G7. Network access to Microsoft cloud endpoints

**Description:** Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

**The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.**

## 3. Implementation Guide — BitLocker Drive Encryption

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A disk encryption policy requiring XTS-AES 256, Entra key escrow, and silent enablement, assigned to a pilot group.*

### Step 1 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `BitLocker-Pilot` → add pilot devices/users. **Create**.

### Step 2 — Create the disk encryption policy

1. Intune → **Endpoint security** → **Disk encryption** → **Create policy**.
2. **Platform:** Windows. **Profile: BitLocker**. **Create**.
3. **Name:** `BitLocker - Pilot`. **Next**.

### Step 3 — Configure BitLocker settings

On **Configuration settings**:

1. **BitLocker – Base Settings**

• **Enable full disk encryption for OS and fixed data drives** = **Yes**.

• **Require storage cards to be encrypted (mobile only)** = not applicable.

• **Hide prompt about third-party encryption** = **Yes**.

• **Allow standard users to enable encryption during Autopilot** = **Yes**.

• **Configure client-driven recovery password rotation** = **Enabled for Microsoft Entra joined devices**.

1. **BitLocker – Fixed Drive Settings**

• **Fixed drive recovery** = **Configure** → **Save BitLocker recovery information to Microsoft Entra ID** = **Yes**; **Store recovery information before enabling** = **Yes**.

• **Block write access to fixed drives not protected by BitLocker** = optional.

1. **BitLocker – OS Drive Settings**

• **Encryption for OS drives** = **Require**.

• **BitLocker with non-compatible TPM chip** = block (or allow PIN if you support non-TPM).

• **OS drive recovery** = **Configure** → **Save BitLocker recovery information to Microsoft Entra ID** = **Yes**; **Store recovery information before enabling** = **Yes**; **Hide recovery options during BitLocker setup** = **Yes**.

• **Encryption method** = **XTS-AES 256-bit**.

1. **BitLocker – Removable Drive Settings** = as required. Click **Next**.

## Step 4 — Assign to the pilot group

1. **Assignments** → **Add groups** → `BitLocker-Pilot`. **Next** → **Create**.

## Step 5 — Apply on a pilot device

1. On a pilot device, sync policy (**Settings** → **Accounts** → **Access work or school** → **Info** → **Sync**).
2. On TPM devices encryption begins **silently**; otherwise the user is prompted for a startup PIN.

## Step 6 — Validate the pilot

1. Run `BITLK-Validation.ps1` → confirms the OS drive is encrypting/encrypted with the right method and a recovery protector exists.
2. In Intune → device → **Recovery keys**, confirm the **recovery key is escrowed**.
3. Run `BITLK-TenantReport.ps1` → confirms encryption status across the pilot group. Complete the **Testing Guide**.

## Step 7 — Expand to production

1. Add a production group to the policy **Assignments**; roll out in waves.
2. Confirm key escrow for each wave (a device that encrypts without an escrowed key is a recovery risk).
3. Re-validate with `BITLK-Validation.ps1` / `BITLK-TenantReport.ps1`.

### *Order of operations*

```
Pilot group → Disk encryption policy → Base/Fixed/OS settings (XTS-AES 256 + Entra escrow + silent) → Assign → Apply → Validate encryption + escrow → Expand
```

# 5. Testing Guide — BitLocker Drive Encryption

Run after the pilot policy applies (Implementation Step 6) and after each production wave.

### Test 1 — Policy applied

**Do:** Intune → **Endpoint security** → **Disk encryption** → `BitLocker - Pilot` → **Device status**. **Expected:** Pilot devices show **Succeeded** (no Error).

### Test 2 — OS drive is encrypted

**Do:** On a pilot device run `BITLK-Validation.ps1` or `manage-bde -status`. **Expected:** OS drive **Protection On**, **Percentage Encrypted = 100%**, **Encryption Method = XTS-AES 256**.

### Test 3 — Silent enablement (TPM devices)

**Do:** Observe a TPM device after policy sync — no user prompt. **Expected:** Encryption begins automatically with no user interaction.

### Test 4 — Recovery key escrowed to Entra

**Do:** Intune → device → **Recovery keys** (or Entra → device → **BitLocker keys**). **Expected:** A recovery key is present for the device.

### Test 5 — Recovery key retrieval works

**Do:** Run `BITLK-Get-RecoveryKey.ps1 -DeviceName <name>` (Graph). **Expected:** The recovery password is returned for an authorized admin.

### Test 6 — Compliance reflects encryption

**Do:** Intune → device → **Compliance** (with a "require encryption" compliance rule). **Expected:** The encrypted device is **Compliant** for the encryption requirement.

### Test 7 — Tenant report

**Do:** Run `BITLK-TenantReport.ps1`. **Expected:** A breakdown of encrypted / encrypting / not-encrypted devices.

### Pass/fail gate

| Gate | Requirement |
|---|---|
| **Pilot pass** | Tests 1–5 pass for the pilot group (escrow confirmed for every device) |
| **Wave pass** | Tests 2, 4, 7 pass before the next wave |

## 6. Troubleshooting Guide — BitLocker Drive Encryption

Use `BITLK-Validation.ps1` / `BITLK-Discovery.ps1` for device state and `BITLK-Remediation.ps1` for safe fixes.

## A. Encryption doesn't start

| Symptom | Cause | Remediation |
|---|---|---|
| Policy "Succeeded" but not encrypting | No TPM, or TPM not ready | Run `BITLK-Discovery.ps1`; enable TPM in UEFI; for non-TPM, configure a startup PIN |
| "Encryptable volume" error | Secure Boot/UEFI or pre-provisioning state | Ensure UEFI + Secure Boot; reboot; re-sync policy |
| Standard user can't enable | "Allow standard users to enable during Autopilot" not set | Enable that base setting |

## B. Recovery key not escrowed (critical)

| Symptom | Cause | Remediation |
|---|---|---|
| Encrypted but no key in Entra | "Store recovery info before enabling" off, or device not Entra joined | Set escrow-before-enable; confirm `AzureAdJoined : YES`; run `BITLK-Remediation.ps1` to back up the key |
| Key missing for older devices | Encrypted before policy | `BITLK-Remediation.ps1` backs up the existing recovery password to Entra |

## C. Wrong encryption method

- Device shows AES-128 / different method → it was encrypted before the policy. Decrypt and re-encrypt with XTS-AES 256, or accept per risk; new devices follow the policy.

## D. TPM problems

- `Get-Tpm` shows not ready → enable/clear TPM in UEFI (coordinate: clearing TPM affects existing keys — have the recovery key first).
- TPM "reduced functionality" → clear via `tpm.msc` (with recovery key in hand).

## E. Diagnostics quick reference

```
manage-bde -status                      # encryption %, method, protection state
(Get-BitLockerVolume -MountPoint C:).KeyProtector   # protectors (TPM, RecoveryPassword)
Get-Tpm                                 # TPM readiness
manage-bde -protectors -get C:          # view recovery password ID
```

Back up an existing key to Entra: `BackupToAAD-BitLockerKeyProtector -MountPoint C: -KeyProtectorId <id>`.

### F. When to escalate

If a device has a TPM ready, is Entra joined, the policy shows Succeeded, and encryption still won't start or the key won't escrow after remediation, collect `manage-bde -status` output and the BitLocker event logs and open a Microsoft support case.

## 7. Best Practices — BitLocker Drive Encryption

- **Escrow the key before enabling.** Always set "Store recovery information before enabling" — a device that encrypts without an escrowed key is a recovery disaster waiting to happen.
- **Pilot first.** Confirm silent enablement and key escrow on a small group before fleet rollout.
- **Require XTS-AES 256.** The modern, strong method for fixed internal drives.
- **Prefer silent enablement on TPM devices.** No user friction; reserve startup PINs for high-security scopes or non-TPM hardware.
- **Enable client-driven key rotation** for Entra-joined devices so recovery passwords refresh after use.
- **Pair with compliance.** Use a "require encryption" compliance rule so unencrypted devices are flagged (and blocked via Conditional Access).
- **Mind the TPM before clearing it.** Clearing/resetting a TPM can invalidate keys — always have the recovery key first.
- **Encrypt fixed data drives too**, with auto-unlock, for full coverage.
- **Report continuously.** Track encryption coverage and escrow with `BITLK-TenantReport.ps1`.
- **Document recovery procedure.** Make sure the help desk knows how to retrieve a recovery key (`BITLK-Get-RecoveryKey.ps1` / Intune device → Recovery keys).

## 8. Deliverables Summary — BitLocker Drive Encryption

### Documentation

| Document | Purpose |
|---|---|
| Executive Summary | Business case and what's in the box |
| 1. Solution Overview | What Intune-managed BitLocker is, how it works, benefits, limitations |
| 2. Requirements Guide | Every prerequisite with Verify / Implement-if-missing |
| Global Requirements (Intune Solutions) | Category-wide prerequisites, published once |
| 3. Implementation Guide | Exact, ordered, click-by-click Intune steps |
| 5. Testing Guide | Validation procedures with pass/fail gates |
| 6. Troubleshooting Guide | Field issues and remediation |
| 7. Best Practices | Field-proven recommendations |

## PowerShell scripts

| Script | What it does |
|---|---|
| `BITLK-Discovery.ps1` | Readiness: TPM, Windows edition, UEFI/Secure Boot, current encryption state |
| `BITLK-Validation.ps1` | Confirms the OS drive is encrypted (XTS-AES 256) with a recovery protector |
| `BITLK-TenantReport.ps1` | Tenant-wide encryption status + recovery-key presence (Microsoft Graph) |
| `BITLK-Remediation.ps1` | Safe remediation; enables encryption and backs up the recovery key to Entra |
| `BITLK-Get-RecoveryKey.ps1` | Retrieves a device's BitLocker recovery key for authorized admins (Microsoft Graph) |

## What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

## Outcome

Fleet-wide full-disk encryption with escrowed recovery keys — device-loss protection and a compliance cornerstone.

# Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

### BITLK-Discovery.ps1

### BITLK-Get-RecoveryKey.ps1

### BITLK-Remediation.ps1

### BITLK-TenantReport.ps1

### BITLK-Validation.ps1